

SHAW AFB, SC, USA

WMO: 747900

Lat: 33.967N

Lon: 80.467W

Elev: 241

StdP: 14.57

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13849

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.3	27.0	5.9	7.5	32.4	10.6	9.5	35.4	23.5	53.9	20.7	52.5	5.4	30	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	96.2	75.7	93.8	75.5	91.5	75.0	79.1	90.1	78.1	88.6	77.3	87.3	8.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.4	138.9	83.1	75.3	133.8	81.9	74.6	130.5	81.2	42.6	90.1	41.6	88.9	40.7	87.3	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.8	17.5	15.6	DB	15.9	99.5	3.4	3.1	13.4	101.7	11.4	103.5	9.5	105.2	7.0	107.4
			WB	14.6	81.1	3.6	1.2	12.0	82.0	9.8	82.7	7.8	83.3	5.2	84.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.4	45.4	49.4	56.0	64.1	72.2	78.8	81.4	80.3	75.2	65.3	54.5	49.0
	DBStd	14.63	10.23	9.40	9.05	7.31	6.24	4.78	3.58	4.01	5.21	7.35	8.13	9.17
	HDD50	560	214	118	47	3	0	0	0	0	0	1	44	133
	HDD65	2390	608	443	301	104	16	1	0	0	3	88	324	502
	CDD50	5808	73	100	233	426	689	864	972	938	757	474	181	101
	CDD65	2163	2	5	22	77	240	414	507	473	310	96	11	6
	CDH74	20296	12	42	235	775	2209	4006	5136	4553	2489	737	79	23
	CDH80	8587	0	3	36	199	846	1834	2446	2081	965	173	4	0
(14) Wind	WSAvg	7.0	7.4	7.8	8.1	8.1	7.4	6.7	6.5	6.0	6.6	6.3	6.4	6.8
(15) Precipitation	PrecAvg	47.0	3.8	3.5	3.8	3.1	3.5	5.2	5.0	5.4	3.9	3.5	2.7	3.4
	PrecMax	59.9	8.1	7.3	7.3	6.5	7.1	10.4	12.0	9.6	8.1	14.7	7.0	6.5
	PrecMin	35.6	1.3	1.3	0.6	0.9	1.3	1.8	2.3	2.6	0.2	0.0	0.2	0.8
	PrecStd	7.7	2.0	1.6	1.7	1.5	1.5	2.1	2.4	2.2	1.9	3.0	1.8	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.4	79.2	83.9	88.2	95.0	98.7	99.1	99.2	94.2	88.1	80.6	77.2
		MCWB	63.0	64.1	64.9	68.0	72.0	75.4	75.6	76.4	73.2	73.6	67.5	66.0
	2%	DB	70.3	74.2	80.4	85.1	90.8	94.9	96.0	95.3	91.2	84.3	76.5	72.6
		MCWB	60.7	61.4	63.3	66.4	71.1	75.3	76.3	76.6	73.5	70.3	63.9	63.9
	5%	DB	66.1	70.1	76.6	82.0	88.1	92.2	93.3	92.7	88.8	81.5	72.6	68.5
		MCWB	58.7	58.3	61.2	65.2	70.5	74.7	76.1	76.2	73.1	68.2	62.0	62.3
	10%	DB	62.2	65.6	72.1	78.8	85.0	89.8	91.0	90.1	86.1	78.3	68.9	64.4
		MCWB	56.2	56.6	59.3	63.7	69.6	74.2	76.1	75.7	72.3	66.7	61.3	59.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.7	66.8	68.3	72.3	75.6	79.5	80.9	80.4	78.0	76.2	70.4	69.5
		MCDB	70.2	74.3	76.7	81.2	85.9	91.2	92.5	92.1	86.9	82.3	74.5	72.6
	2%	WB	63.6	64.4	66.2	70.1	74.0	77.8	79.2	79.1	76.5	74.1	67.8	66.3
		MCDB	67.8	71.4	74.9	78.5	84.6	89.6	90.2	90.0	85.1	80.4	71.9	70.2
	5%	WB	60.6	61.6	64.3	68.4	72.8	76.7	78.1	78.1	75.4	72.1	65.7	63.7
		MCDB	64.8	67.4	71.9	77.0	83.0	87.9	88.9	88.1	83.6	77.7	70.5	66.6
	10%	WB	57.0	58.4	62.2	66.6	71.5	75.6	77.3	77.2	74.3	69.7	62.9	60.4
		MCDB	61.3	64.0	69.0	75.1	80.7	85.8	87.5	86.3	81.6	75.6	67.7	64.7
(35) Mean Daily Temperature Range	5% DB	MDBR	20.8	21.9	23.4	23.3	21.5	19.5	18.9	18.6	18.7	21.2	22.3	20.4
		MCDBR	24.6	26.9	27.7	26.6	24.2	22.2	21.6	21.6	21.9	23.3	25.3	23.8
	5% WB	MCWBR	16.0	16.3	13.4	11.3	8.5	7.2	6.8	6.4	7.5	10.9	14.8	15.8
		MCWBR	20.9	23.9	22.9	21.8	20.4	20.0	19.3	19.1	17.9	18.0	19.9	20.2
(40) Clear-Sky Solar Irradiance	taub		0.320	0.329	0.364	0.409	0.473	0.505	0.552	0.521	0.438	0.394	0.344	0.316
		taud	2.499	2.471	2.398	2.262	2.131	2.085	1.978	2.056	2.275	2.365	2.466	2.548
	Ebn at Noon		283	292	289	279	261	252	239	245	261	265	271	279
		Edh at Noon	27	31	36	43	49	52	57	52	40	34	28	24
(44) All-Sky Solar Radiation	RadAvg		882	1083	1427	1796	1951	2030	1935	1753	1507	1262	969	775
	RadStd		46	113	108	128	152	141	121	87	129	152	75	69

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.88	N/A	N/A	N/A	N/A	N/A	N/A	+319	+241
(47) Regional (64 neighbors)		+0.99	N/A	N/A	N/A	N/A	N/A	N/A	+342	+243

Nomenclature: See separate page